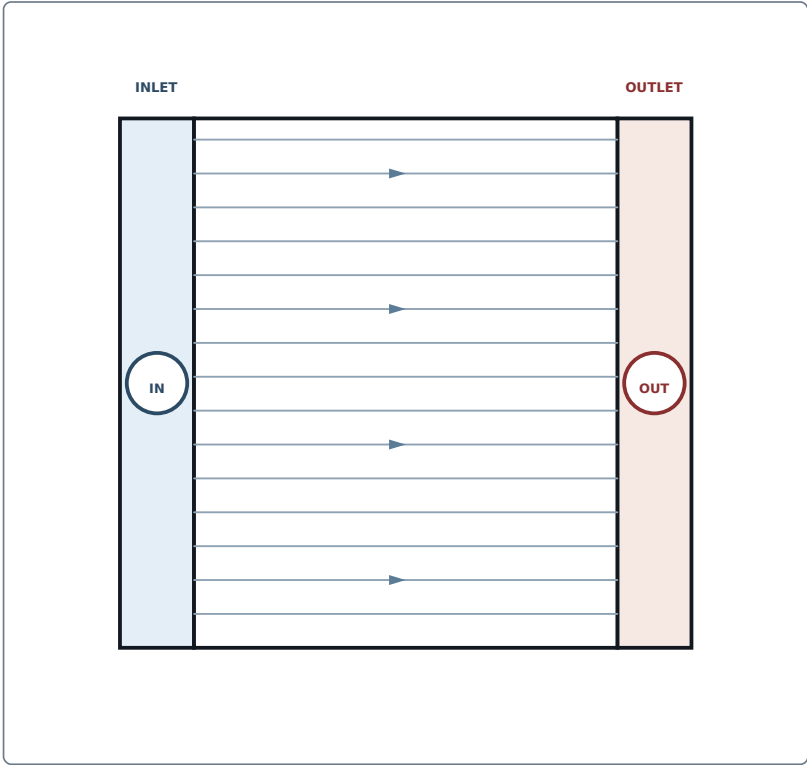


FLOW PATH

SCALE 1.4:1



125 PARALLEL CHANNELS — 0.25 W x 4.0 H x 40 LONG

Single pass, straight. No U-turn, no serpentine:

DESIGN POINT — 1.5 L/min, 700 W		
PARAMETER	WATER	PG25
velocity in channel	0.20 m/s	0.20 m/s
Reynolds number	170	71
heat transfer coefficient	7641 W/m2K	5300 W/m2K
fin efficiency	0.47	0.55
coolant temperature rise	6.8 K	7.0 K
pressure drop, plate only	2.5 kPa	6.1 kPa

ROBUSTNESS — CONSTANT FLOW					
CHANNEL BLOCKAGE	OPEN	VELOCITY	R total	DIE @45 C	dP
all clear	125	0.20 m/s	0.0295	65.7 C	6.1 kPa
5 % blocked	118	0.21 m/s	0.0299	66.0 C	6.5 kPa
10 % blocked	112	0.22 m/s	0.0303	66.2 C	6.9 kPa
20 % blocked	100	0.25 m/s	0.0313	66.9 C	7.7 kPa

READING THESE NUMBERS

- PG25 is the design fluid. Water is shown only to bound the gain from inhibited water where freeze protection is not required — about 5 % on total resistance.
- A SINGLE blocked channel is not a thermal event. The 1.5 mm copper base carries its ~10 W sideways 0.4 mm to the neighbours for roughly 0.1 K.
- Progressive fouling across many channels is the real risk, and it shows up first as rising pressure drop, not as a rising die temperature.
- So instrument the loop for DIFFERENTIAL PRESSURE, not only temperature, and filter to 50 um at the CDU. Service trigger: dP 20 % above the commissioning baseline at the same flow.

FLOW SENSITIVITY — PG25, 700 W				
FLOW	R total	DIE @45 C	RISE	dP
0.75 L/min	0.0373	71.1 C	14.0 K	3.1 kPa
1.00 L/min	0.0333	68.3 C	10.5 K	4.1 kPa
1.50 L/min	0.0295	65.7 C	7.0 K	6.1 kPa
2.00 L/min	0.0278	64.4 C	5.2 K	8.2 kPa
3.00 L/min	0.0261	63.3 C	3.5 K	12.3 kPa

Below 1.0 L/min the plate loses ground quickly. Above 2.0 L/min it buys almost nothing — spend the pump head elsewhere in the loop.

INDIA-VC THERMAL HARDWARE					ISO 128 / ISO 2768-mK
TITLE THERMAL-HYDRAULIC SPECIFICATION H100 SXM5 700 W / PG25 / DESIGN POINT AND SENSITIVITIES			SCALE NONE	UNITS mm	THIRD ANGLE PROJECTION
DRAWING No. CP-103			REV A	SHEET 4 / 6	
DRAWN AI-ASSISTED DRAFT	CHECKED — PENDING —	MATERIAL SEE BOM	FINISH SEE NOTES	DATE 2026-09-10	